

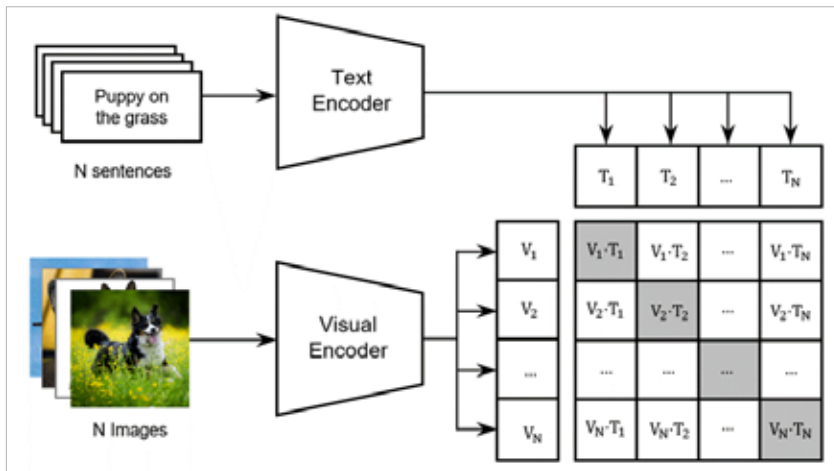


Figure 7: Contrastive language-image pretraining (CLIP)

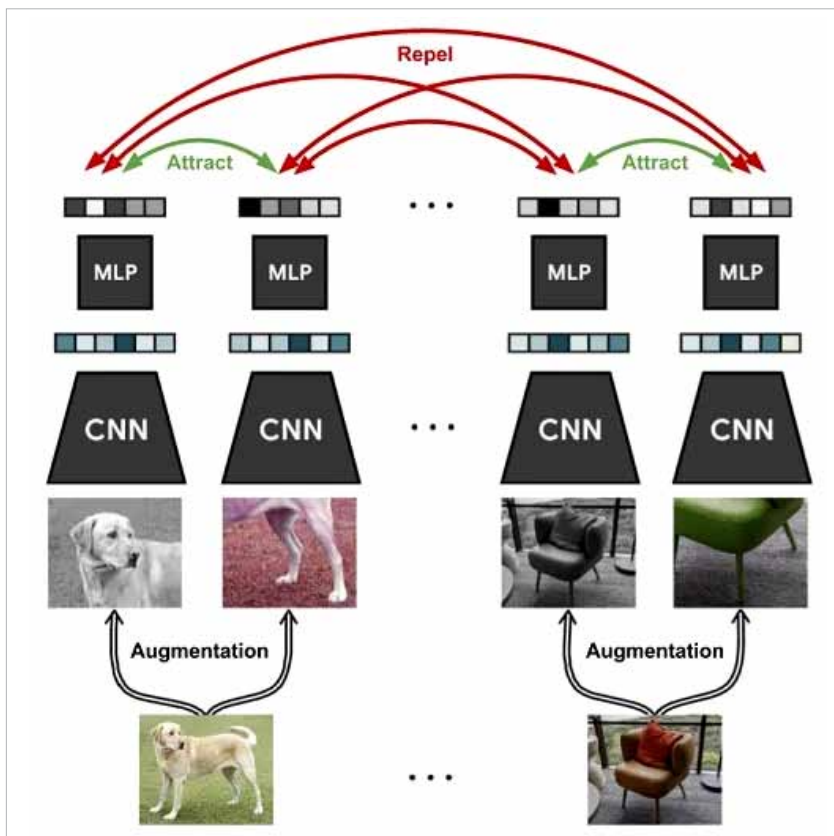


Figure 8: SimCLR architecture

while minimising agreement between dissimilar ones. However, what sets MoCo apart is its integration of a dynamic memory bank, which serves as a source of negative samples. By maintaining a queue of previously generated representations, MoCo

enables models to learn from a broader context and capture more diverse features.

MoCo incorporates a dynamic memory bank and a momentum update mechanism. This momentum update facilitates a more stable and robust

learning process, allowing the model to accumulate knowledge from previous iterations. Consequently, MoCo enhances the representation learning process by incorporating a broader range of negative samples through its memory bank. One of the significant advantages of MoCo is its scalability. By leveraging a large memory bank, MoCo can efficiently store and utilise negative samples from a vast unlabelled data set. This scalability leads to more comprehensive and diverse representation learning, enabling models to capture richer and more discriminative features.

## Applications of the latest models

I have briefly explained the most trending research topics in the visual data analysis field. We have also seen applications of CNNs, RNNs, and other deep learning models in healthcare, security, etc. Yet, the question remains: how will these newer models replace traditional techniques?

**Vision transformers:** Vision transformers can replace traditional CNNs and feature extraction methods in image classification, object detection, and segmentation tasks. They offer improved performance in understanding and analysing images.

**CLIP:** CLIP can replace traditional techniques in tasks like image classification, image retrieval, and multimodal tasks such as image captioning and visual question answering. It combines vision and language understanding, enabling models to comprehend and generate information from textual and visual inputs.

**SimCLR:** SimCLR can replace traditional techniques in representation learning tasks. It learns meaningful representations from unlabelled data, enhancing image classification and transfer learning applications.

**MoCo:** MoCo enhances representation learning and can replace